

Simulate Responses of Estimated VARX Model

This example shows how to estimate a multivariate time series model that contains lagged endogenous and exogenous variables, and how to simulate responses. The response series are the quarterly:

[Open Script](#)

- Changes in real gross domestic product (rGDP) rates (y_{1t})
- Real money supply (rM1SL) rates (y_{2t})
- Short-term interest rates (i.e., three-month treasury bill yield, y_{3t})

from March, 1959 through March, 2009 . The exogenous series is the quarterly changes in the unemployment rates (x_t).

Suppose that a model for the responses is this VARX(4,3) model

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \end{bmatrix} + \begin{bmatrix} x_t & 0 & 0 \\ 0 & x_t & 0 \\ 0 & 0 & x_t \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} + \begin{bmatrix} \phi_{11}^{(1)} & \phi_{12}^{(1)} & \phi_{13}^{(1)} \\ \phi_{21}^{(1)} & \phi_{22}^{(1)} & \phi_{23}^{(1)} \\ \phi_{31}^{(1)} & \phi_{32}^{(1)} & \phi_{33}^{(1)} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ y_{3,t-1} \end{bmatrix} + \dots + \begin{bmatrix} \phi_{11}^{(4)} & \phi_{12}^{(4)} & \phi_{13}^{(4)} \\ \phi_{21}^{(4)} & \phi_{22}^{(4)} & \phi_{23}^{(4)} \\ \phi_{31}^{(4)} & \phi_{32}^{(4)} & \phi_{33}^{(4)} \end{bmatrix} \begin{bmatrix} y_{1,t-4} \\ y_{2,t-4} \\ y_{3,t-4} \end{bmatrix}$$

Preprocess the Data

Load the U.S. macroeconomic data set. Flag the series and their periods that contain missing values (indicated by NaN values).

```
load Data_USEconModel
varNaN = any(ismissing(DataTable),1); % Variables containing NaN values
seriesWithNaNs = series(varNaN)

seriesWithNaNs =

1x5 cell array

Columns 1 through 3

'(FEDFUNDS) Effec...'    '(GS10) Ten-year ...'    '(M1SL) M1 money ...'

Columns 4 through 5

'(M2SL) M2 money ...'    '(UNRATE) Unemplo...'
```

In this data set, the variables that contain missing values entered the sample later than the other variables. There are no missing values after sampling started for a particular variable.

vgxvarx accommodates missing values for responses, but not for regression variables. Flag all periods corresponding to a missing regression variable value.

```
idx = ~isnan(DataTable.UNRATE);
```

For the rest of the example, consider only those values that of the series indicated by a true in idx.

Compute rGDP and rM1SL, and the growth rates of rGDP, rM1SL, short-term interest rates, and the unemployment rate. Description contains a description of the data and the variable names. Reserve the last three years of data to investigate the out-of-sample performance of the estimated model.

```

rGDP = DataTable.GDP(idx)./(DataTable.GDPDEF(idx)/100);
rM1SL = DataTable.M1SL(idx)./(DataTable.GDPDEF(idx)/100);

dLRGDP = diff(log(rGDP));           % rGDP growth rate
dLRM1SL = diff(log(rM1SL));          % rM1SL growth rate
d3MTB = diff(DataTable.TB3MS(idx));  % Change in short-term interest rate (3MTB)
dUNRATE = diff(DataTable.UNRATE(idx)); % Change in unemployment rate

T = numel(d3MTB);    % Total sample size
oosT = 12;           % Out-of-sample size
estT = T - oosT;      % Estimation sample size
estIdx = 1:estT;      % Estimation sample indices
oosIdx = (T - 11):T;  % Out-of-sample indices
dates = dates((end - T + 1):end);

EstY = [dLRGDP(estIdx) dLRM1SL(estIdx) d3MTB(estIdx)]; % In-sample responses
estX = dUNRATE(estIdx);                               % In-sample exogenous data
n = size(EstY,2);

OOSY = [dLRGDP(oosIdx) dLRM1SL(oosIdx) d3MTB(oosIdx)]; % Out-of-sample responses
oosX = dUNRATE(oosIdx);                               % Out-of-sample exogenous data

```

Create the Design Matrices

Create an `estT`-by-1 cell vector of block design matrices that associate the predictor series with each response such that the responses do not share a coefficient.

```

EstExpandX = kron(estX,eye(n));
EstCellX = mat2cell(EstExpandX,n*ones(estT,1));
nX = size(EstExpandX,2);

```

Specify the VARX Model

Specify a multivariate time series model object that characterizes the VARX(4,3) model using `vgxset`.

```
Mdl = vgxset('n',n,'nAR',4,'nX',nX,'Constant',true);
```

Estimate the VAR(4) Model

Estimate the parameters of the VARX(4,3) model using `vgxvarx`. Display the parameter estimates.

```

EstMdl = vgxvarx(Mdl,EstY,EstCellX);
vgxdisp(EstMdl)

```

Model : 3-D VARMAX(4,0,3) with Additive Constant
 Conditional mean is AR-stable and is MA-invertible

a Constant:

0.00811792
 0.000709263
 0.0465824

b Regression Parameter:

-0.0162116
 -0.00163933
 -1.50115

AR(1) Autoregression Matrix:

-0.0375772 -0.0133236 0.00108218
 -0.00519697 0.177963 -0.00501432
 -0.873992 -6.89049 -0.165888

AR(2) Autoregression Matrix:

0.0753033 0.0775643 -0.001049
 0.00282857 0.29064 -0.00159574
 4.00724 0.465046 -0.221024

AR(3) Autoregression Matrix:

-0.0927688 -0.0240239 -0.000549057
 0.0627837 0.0686179 -0.00212185
 -7.52241 10.247 0.227121

AR(4) Autoregression Matrix:

0.0646951 -0.0792765 -0.000176166
 0.0276958 0.00922231 -0.000183861
 1.38523 -11.8774 0.0518154

Q Innovations Covariance:

3.57524e-05 7.05807e-06 -4.23542e-06
 7.05807e-06 9.67992e-05 -0.00188786
 -4.23542e-06 -0.00188786 0.777151

EstMdl is a multivariate time series model object containing the estimated parameters.

Simulate Out-Of-Sample Response Paths Using the Same Exogenous Data per Path

Simulate 1000, 3 year response series paths from the estimated model assuming that the exogenous unemployment rate is a fixed series. Since the model contains 4 lags per endogenous variable, specify the last 4 observations in the estimation sample as presample data.

```
OOSExpandX = kron(oosX,eye(n));
OOSCellX = mat2cell(OOSExpandX,n*ones(oosT,1));
numPaths = 1000;
Y0 = EstY((end-3):end,:);
rng(1); % For reproducibility
YSim = vgxsim(EstMdl,oosT,OOSCellX,Y0,[],numPaths);
```

YSim is a 12-by-3-by-1000 numeric array of simulated responses. The rows of YSim correspond to out-of-sample periods, the columns correspond to the response series, and the leaves correspond to paths.

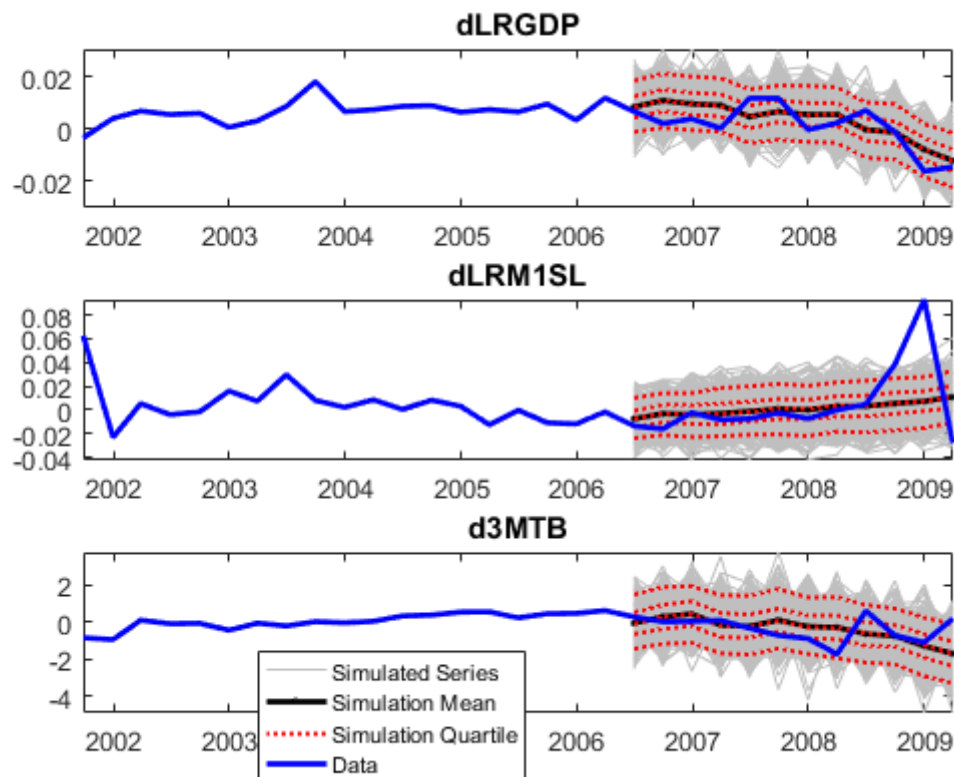
Plot the response data and the simulated responses. Identify the 5%, 25%, 75% and 95% percentiles, and the mean and median of the simulated series at each out-of-sample period.

```

YSimBar = mean(YSim,3);
YSimQrt1 = quantile(YSim,[0.05 0.25 0.5 0.75 0.95],3);
RepDates = repmat(dates(oosIdx),1,1000);
respNames = {'dLRGDP' 'dLRM1SL' 'd3MTB'};

figure;
for j = 1:n;
    subplot(3,1,j);
    h1 = plot(dates(oosIdx),squeeze(YSim(:,j,:)), 'Color',0.75*ones(3,1));
    hold on;
    h2 = plot(dates(oosIdx),YSimBar(:,j), '-k', 'LineWidth',2);
    h3 = plot(dates(oosIdx),squeeze(YSimQrt1(:,j,:)), ':r', 'LineWidth',1.5);
    h4 = plot(dates((end - 30):end),[EstY((end - 18):end,j);OOSY(:,j)],...
        'b', 'LineWidth',2);
    title(sprintf('%s',respNames{j}));
    datetick;
    axis tight;
    hold off;
end
legend([h1(1) h2(1) h3(1) h4],[ 'Simulated Series','Simulation Mean',...
    'Simulation Quartile','Data'],'Location',[0.4 0.1 0.01 0.01],...
    'FontSize',8);

```



Simulate Out-Of-Sample Response Paths Using Random Exogenous Data

Suppose that the change in the unemployment rate is an AR(4) model, and fit the model to the estimation sample data.

```

MdlUNRATE = arima('ARLags',1:4);
EstMdlUNRATE = estimate(MdlUNRATE,estX,'Display','off');

```

EstMdlUNRATE is an [arima](#) model object containing the parameter estimates.

Simulate 1000, 3 year paths from the estimated AR(4) model for the change in unemployment rate. Since the model contains 4 lags, specify the last 4 observations in the estimation sample as presample data.

```
XSim = simulate(EstMdlUNRATE,oosT,'Y0',estX(end-3:end),...
    'NumPaths',numPaths);
```

XSim is a 12-by-1000 numeric matrix of simulated exogenous paths. The rows correspond to periods and the columns correspond to paths.

Create a cell matrix of block design matrices to organize the exogenous data, where each column corresponds to a path.

```
ExpandXSim = kron(XSim,eye(n));
size(ExpandXSim)
CellXSim = mat2cell(ExpandXSim,n*ones(oosT,1),n*ones(1,numPaths));
size(CellXSim)
CellXSim{1,1}
```

ans =

```
36      3000
```

ans =

```
12      1000
```

ans =

```
0.7901      0      0
      0    0.7901      0
      0      0    0.7901
```

ExpandXSim is a 36-by-3000 numeric matrix, and CellXSim is a 12-by-1000 cell matrix of mutually exclusive, neighboring, 3-by-3 block matrices in ExpandXSim.

Simulate 3 years of 1000 future response series paths from the estimated model using the simulated exogenous data. Since the model contains 4 lags per endogenous variable, specify the last 4 observations in the estimation sample as presample data.

```
YSimRX = vgxsim(EstMdl,oosT,CellXSim,Y0,[],numPaths);
```

YSimRX is a 12-by-3-by-1000 numeric array of simulated responses.

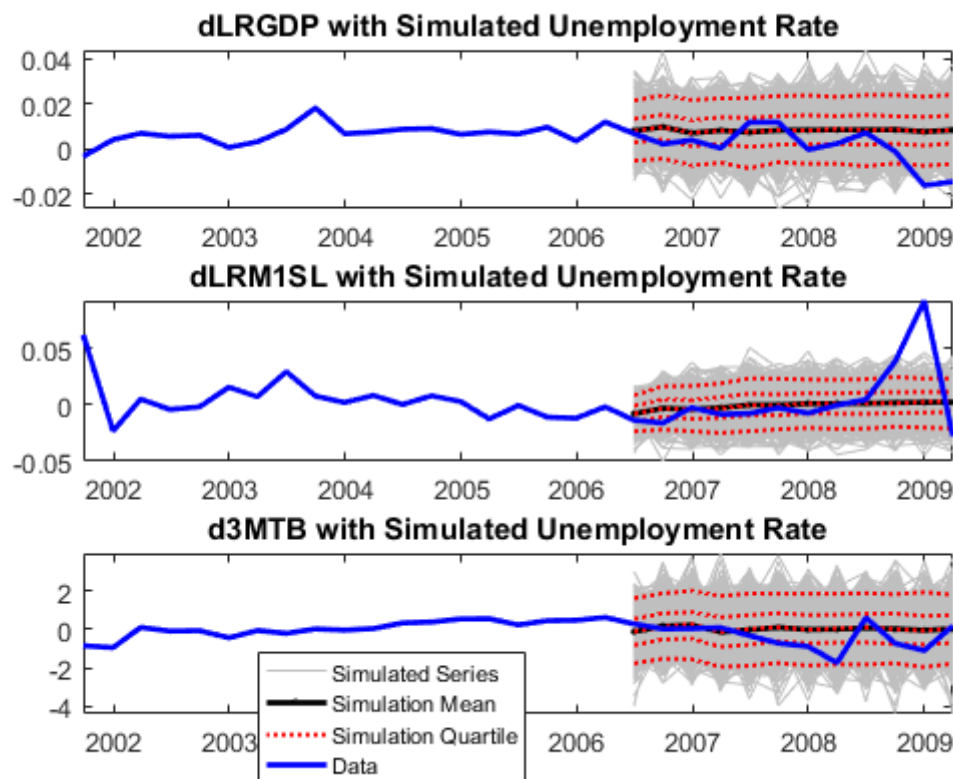
Plot the response data and the simulated responses. Identify the 5%, 25%, 75% and 95% percentiles, and the mean and median of the simulated series at each out-of-sample period.

```

YSimBarRX = mean(YSimRX,3);
YSimQrtlRX = quantile(YSimRX,[0.05 0.25 0.5 0.75 0.95],3);

figure;
for j = 1:n;
    subplot(3,1,j);
    h1 = plot(dates(oosIdx),squeeze(YSimRX(:,j,:)), 'Color',0.75*ones(3,1));
    hold on;
    h2 = plot(dates(oosIdx),YSimBarRX(:,j), '-k', 'LineWidth',2);
    h3 = plot(dates(oosIdx),squeeze(YSimQrtlRX(:,j,:)), 'r', 'LineWidth',1.5);
    h4 = plot(dates((end - 30):end),[EstY((end - 18):end,j);OOSY(:,j)],...
        'b', 'LineWidth',2);
    title(sprintf('%s with Simulated Unemployment Rate',respNames{j}));
    datetick;
    axis tight;
    hold off;
end
legend([h1(1) h2(1) h3(1) h4],[ 'Simulated Series','Simulation Mean',...
    'Simulation Quartile','Data'],'Location',[0.4 0.1 0.01 0.01],...
    'FontSize',8)

```



See Also

[mvregress](#) | [vgxpred](#) | [vgxvarx](#)

Related Examples

- [Fit a VAR Model](#)
- [Implement Seemingly Unrelated Regression Analyses](#)
- [Estimate the Capital Asset Pricing Model Using SUR](#)

- [VAR Model Case Study](#)

More About

- [Vector Autoregressive \(VAR\) Models](#)
 - [Multivariate Time Series Models with Regression Terms](#)
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